

Ex. No. 07	APPLICATION STATE MANAGEMENT AND SERIALIZATION
Date of Exercise	
Git Hub Host Link	https://github.com/JenithB/flutter.git

Aim

To manage application state and serialize/deserialize data using JSON in a Flutter application.

Description

This experiment demonstrates application state management using `setState()` and JSON serialization/deserialization using a `Student` model. Student data is converted from a Dart object into JSON format using `toJson()` and converted back into a Dart object using `fromJson()`. A list of students is displayed dynamically using `ListView.builder()`.

Procedure:

1. A new Flutter project named `state_app` was created using the Flutter command.
2. A Dart file named `student.dart` was created inside the `lib` folder to define the `Student` model.
3. The `Student` class was created with the properties `id`, `name`, and `course`.
4. The `toJson()` method was implemented to convert the `Student` object into a JSON-compatible `Map`.
5. The `fromJson()` factory constructor was implemented to convert JSON data back into a `Student` object.
6. A `Student` object was created with the values 1, Ragu, and CSE.
7. The `toJson()` method was used to serialize the `Student` object into JSON data.
8. The `fromJson()` method was used to deserialize the JSON data back into a `Student` object.
9. A `List<String>` was created to maintain the student names as application state.
10. The `setState()` method was used to add a student to the list whenever the Add Student button was pressed.
11. `ListView.builder()` was used to dynamically display the student names.
12. The application was executed and tested in the Chrome browser.

Program**student.dart**

```
class Student {  
  
  final int id;  
  
  final String name;  
  
  final String course;  
  
  Student({  
  
    required this.id,  
  
    required this.name,  
  
    required this.course,  
  
  });  
  
  Map<String, dynamic> toJson() {  
  
    return {  
  
      'id': id,  
  
      'name': name,  
  
      'course': course,  
  
    };  
  
  }  
  
  factory Student.fromJson(Map<String, dynamic> json) {  
  
    return Student(  
  
      id: json['id'],  
  
      name: json['name'],  
  
      course: json['course'],  
  
    );  
  
  }  
}
```

```
);  
  
}  
  
}
```

main.dart

```
import 'package:flutter/material.dart';  
  
import 'student.dart';  
  
  
void main() {  
  
  runApp(const StateApp());  
  
}  
  
  
class StateApp extends StatelessWidget {  
  
  const StateApp({super.key});  
  
  
  @override  
  
  Widget build(BuildContext context) {  
  
    return MaterialApp(  
  
      debugShowCheckedModeBanner: false,  
  
      title: 'State Management',  
  
      home: const StudentPage(),  
  
    );  
  
  }  
  
}
```

```
class StudentPage extends StatefulWidget {

  const StudentPage({super.key});

  @override

  State<StudentPage> createState() => _StudentPageState();

}

class _StudentPageState extends State<StudentPage> {

  List<String> students = [];

  Student student = Student(

    id: 1,

    name: 'Raj',

    course: 'CSE',

  );

  @override

  void initState() {

    super.initState();

    Map<String, dynamic> data = student.toJson();

    print(data);

    Student newStudent = Student.fromJson(data);

    print(newStudent.name);

  }

  void addStudent() {

    setState(() {

      students.add('Raj');

    });

  }

}
```

@override

Widget build(BuildContext context) {

 return Scaffold(

 appBar: AppBar(

 title: const Text('Student State'),

),

 body: Column(

 children: [

 const SizedBox(height: 20),

 ElevatedButton(

 onPressed: addStudent,

 child: const Text('Add Student'),

),

 Expanded(

 child: ListView.builder(

 itemCount: students.length,

 itemBuilder: (context, index) {

 return ListTile(

 title: Text(students[index]),

);

 },

),

),

],

),

Output

Student State & Serialization

Name: Jenith B

Reg No: URK23CS1268

+ Add Student

Result

Thus, the Flutter application was successfully developed to manage changing student data using `setState()`. The Student object was successfully serialized into JSON and deserialized back into a Dart object. The student data was also dynamically displayed using `ListView.builder()`.